Prefrontal-Inspired Control for Adaptive Goal Selection in Reinforcement Learning

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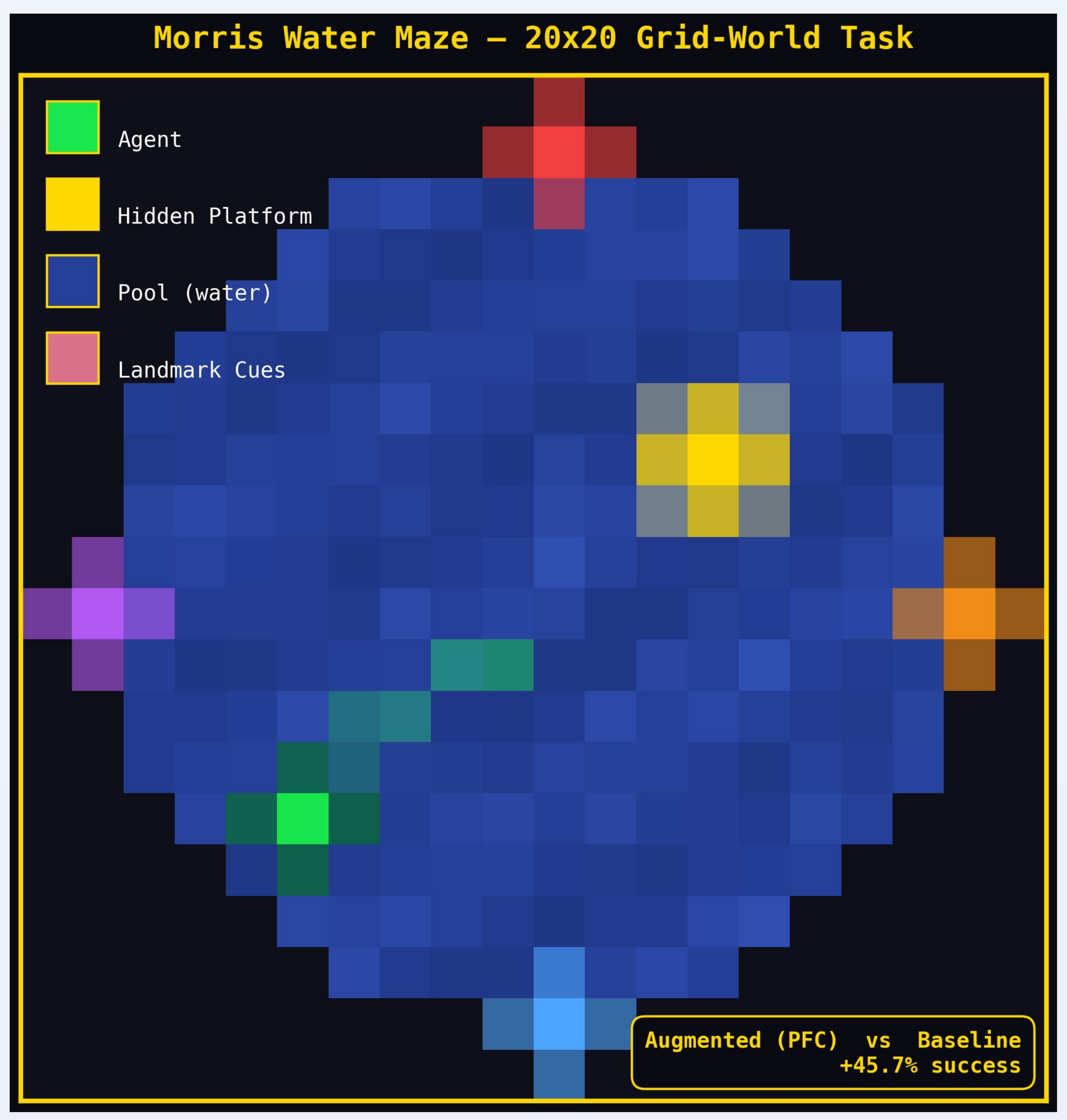
Abstract

- Prefrontal cortex (PFC) must dynamically toggle between three core behavioral modes:
  - Exploration
  - Goal-Directed Approach
  - Exploitation
- Implemented two-level Deep RL agent:
  - Meta-Controller: GRU-based recurrent network that selects sub-objective embeddings
  - Lower-Level Policy: Action policy conditioned on meta-controller's selection, sharing common convolutional encoder
- Augmented agent was tested against a parameter-matched PPO baseline across four grid-world tasks modeled on classical behavioral protocols

Example grid-world task schematic



Methods

Augmented PFC model architecture

Observation → CNN Encoder → z (256)

Meta-controller
GRUCell: every 16 steps, selects k ∈ {E, A, X}

Action Policy
π(a | z, k)
[z; embed(k)] 288-dim → a

Meta loss
`sparse + 0.2r_intr`

Policy loss
`r_task + 0.08r_intr`

Baseline (PPO)

Observation → CNN Encoder → z (256)

Action Policy
π(a | z, k)
[z; embed(k)] 288-dim → a

Policy loss
`r_task + 0.08r_intr`

Sub-objective library

Mode	Intrinsic signal	Biological counterpart
Explore	Novel cell visitation count	Dopaminergic novelty (VTA)
Approach	Decreasing distance to target	Mesolimbic goal-directed
Exploit	Proximity to goal ("harvest")	Dorsal-striatal habit

Task suite

MWM	Explore	Approach	Exploit
Visual foraging	Explore	Approach	Exploit
Dynamic obstacles	Explore	Approach	Exploit
Cued visual search	Explore	Approach	Exploit

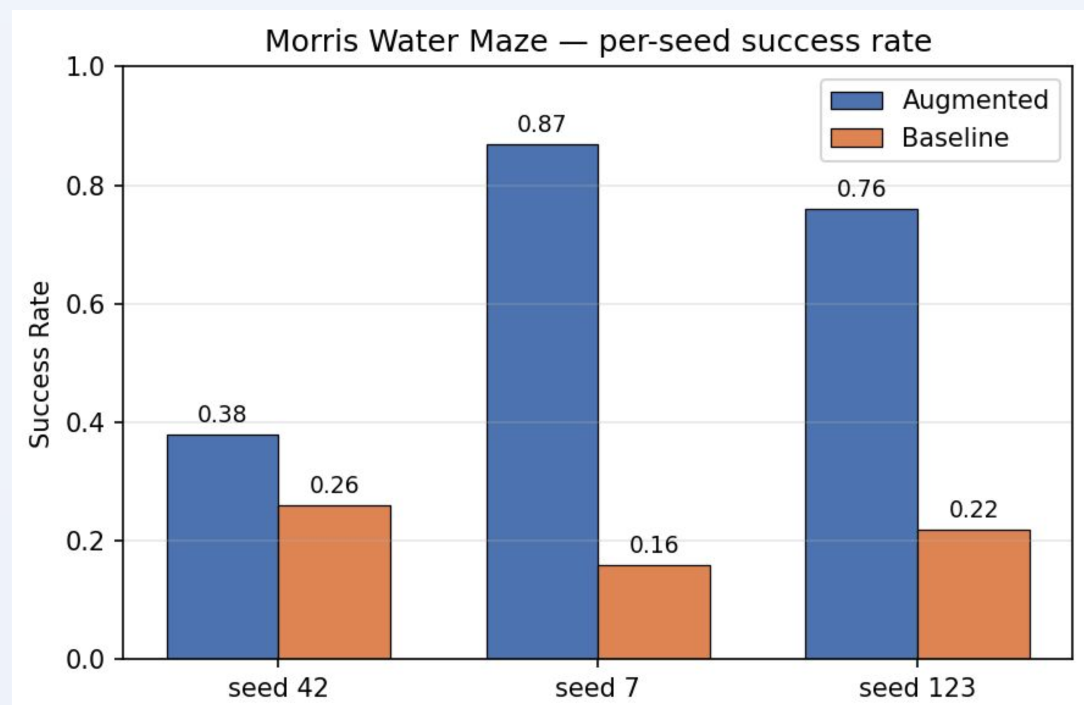
- Temporal commitment of 16 consecutive steps per subobjective, matching timescale of biological PFC executive control
- Fairness guarantees across comparison; sole difference being augmented model's intrinsic reward & suobjective conditioning

Results

+29.5% mean success 3-seed avg	**12/12** task x seed wins overall
+65.6% zero-shot transfer gain	**20 eps** few-shot threshold (aug)

Multi-task success rate (3-seed avg)

Task	Augmented	Baseline	Δ
Morris Water Maze	0.67	0.21	+45.7%
Visual foraging	0.73	0.49	+24.3%
Dynamic obstacles	0.74	0.47	+27.8%
Cued visual search	0.14	0.08	+6.0%
Overall mean	**0.572**	**0.313**	**+25.9%**



Strat distribution/entropy (seed 42)

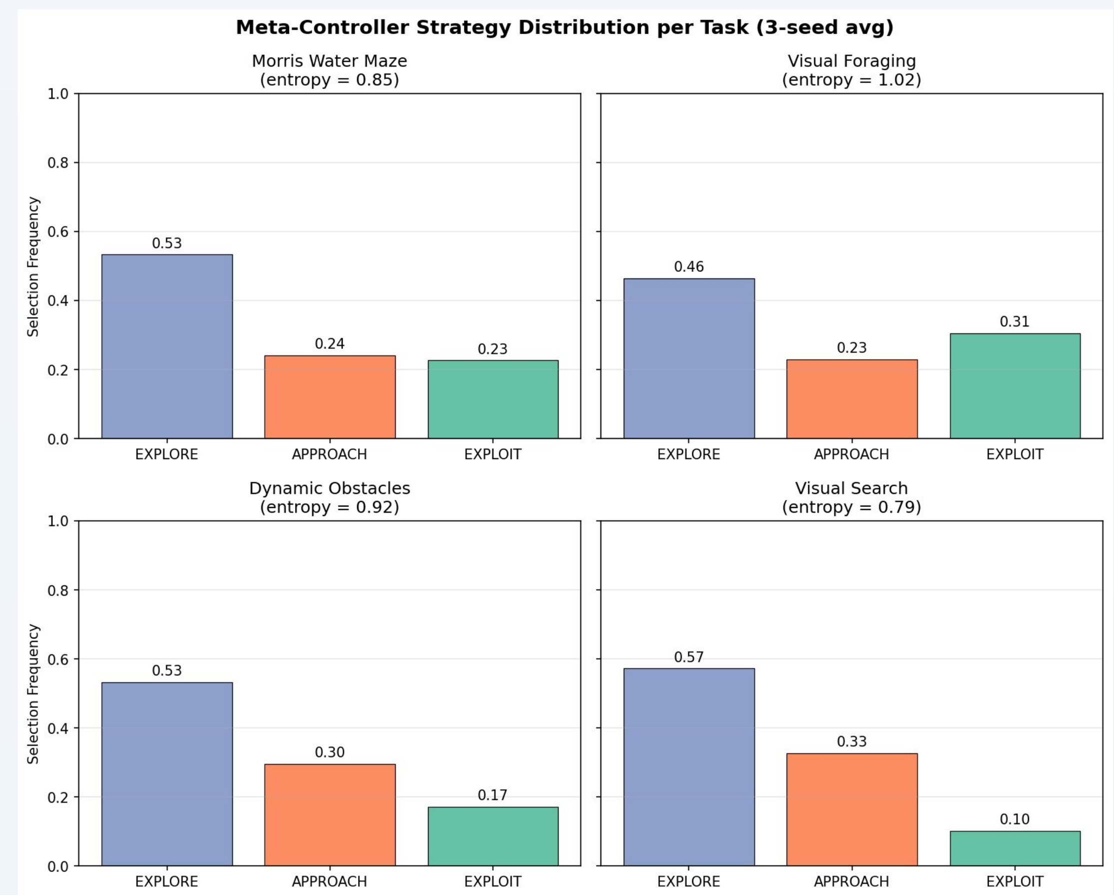
Task	Explore	Approach	Exploit	Harvest
Morris Water Maze	0.30	0.36	0.34	1.10
Visual foraging	0.38	0.38	0.25	1.08
Dynamic obstacles	0.34	0.52	0.14	0.98
Cued visual search	0.29	0.53	0.18	1.00

Mean forgetting

1.25% (seed 42), 3.0% (seeds 7 and 123); forgetting was zero or negative on several task pairs

Discussion

- Augmented model outperformed baseline in **100%** of tasks and seeds, showing a **+25.9%** overall success rate increase.
- High entropy **(0.6-1.1)** confirms the meta-controller avoids mode collapse, maintaining a flexible "behavioral library"
- Results validate the biological model of PFC as a system modulating basic reward circuitry



Next steps

- Learned vs hand-designed, intrinsic reward functions to further extend task coverage, especially for visual search
- Strategy-switching patterns qualitatively mirror PFC dynamics in rodent two-step tasks, but is this mechanistic equivalence?

Acknowledgements

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